

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T07 · Compound Interest and Depreciation

Questions

Student book · questions with writing space

■ 16 pages

FAMILY MAP

Compound Interest Depreciation

F01	Accumulate compound growth	8 Q 30 marks
F02	Accumulate depreciation	2 Q 6 marks
F03	Compare compound and mixed investment paths	3 Q 11 marks
F04	Recover a missing compound parameter	2 Q 6 marks

4MA1-2H Q9

3 marks

- Give your answer correct to the nearest koruna.

..... koruna

Question 26

4MA1-2H Q7 whole

MODULAR UNIT 2

3 marks

- 7 Hermione buys a boat for \$26 800
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest dollar.

\$.....

Question 22

4MA1-2H Q8

MODULAR UNIT 2

3 marks

- 8 Charlotte buys a painting for \$680
The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.
Give your answer correct to the nearest \$

\$.....

4MA1-2H Q9 M01

3 marks

- Give your answer correct to the nearest euro.

..... euros

Question 07

4MA1-1H Q6 Q6(b)

MODULAR UNIT 2

3 marks

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

- (b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %
(3)

4MA1-2H Q9 Q9

3 marks

- dollars

Question 08

4MA1-1H Q11

MODULAR UNIT 2

3 marks

- 11 Himari invests 200 000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and $x\%$ for each of the second year and the third year.

The value of the investment at the end of the third year is 209 754 yen.

Work out the value of x

Give your answer correct to one decimal place.

$x = \dots\dots\dots$

The compound interest rate changes for the third year.
At the end of 3 years, there is a total of \$6311.16 in the account.

Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

(a) Work out the value of Chen's investment at the end of 3 years. Give your answer to the nearest yuan.

..... yuan
(3)

The value of his investment decreased by 6.5% each year.

(b) Work out the value of P .

$P =$

7 Ali and Badia each have 25 000 dollars to invest.

Cyclone Bank	Tornado Bank
Invest 25 000 dollars 4.5% compound interest per year for 3 years	Invest 25 000 dollars Receive 1150 dollars interest each year for 3 years

Ali invests in the Cyclone Bank for 3 years.

Badia invests in the Tornado Bank for 3 years.

By the end of the 3 years, Ali will have received more interest than Badia.

How much more?

Show your working clearly.

Give your answer correct to the nearest dollar.

..... dollars

$x =$

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- 5 Himari's annual salary is 3 130 000 Japanese Yen (JPY).
She gets a salary increase of 4%

(a) Work out Himari's salary after this increase.

.....JPY
(3)

Kaito bought a car.

The value of the car when Kaito bought it was 750 000 JPY.

At the end of each year, the value of his car had depreciated by 15%

- (b) Work out the value of Kaito's car at the end of 3 years.
Give your answer correct to the nearest JPY.

.....JPY
(3)

8 Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

Bank **G**
1.6% per year
compound interest

Bank **H**
2.9% interest added after
two years

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?

..... Swiss francs

November 2025 | 2H | Question 5 | 6 marks

Chapter OL-T07: Compound Interest & Depreciation

5 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.

On Saturday, Otis sells 87 ice creams.

- (a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %
(3)

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

- (b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs
(3)